Questions

Q1.

Figure 10 shows the difference in refraction of sound waves and light waves when these waves travel from air into water.

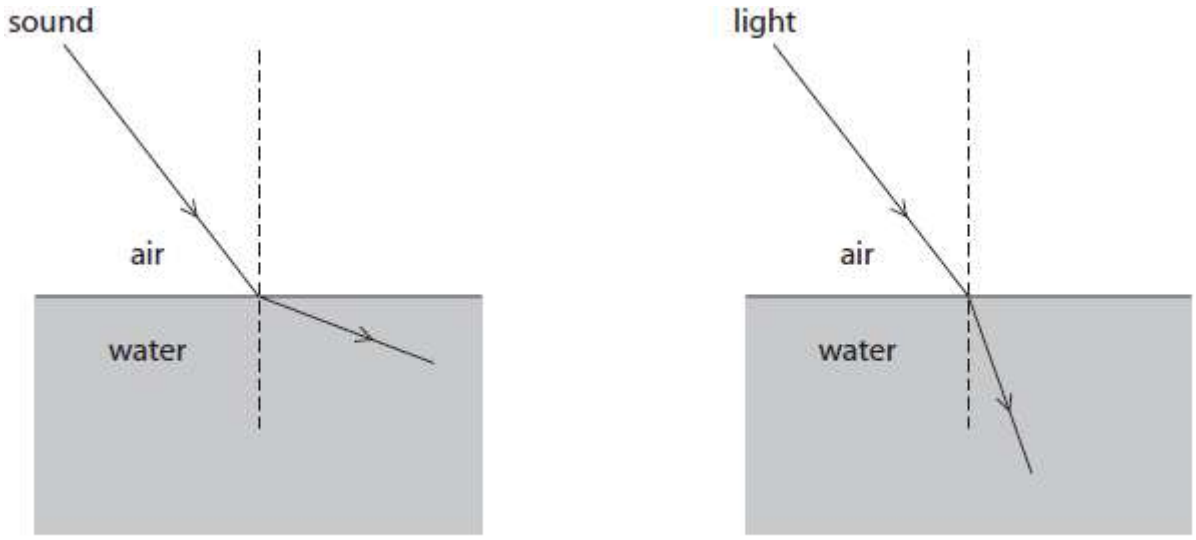

Figure 10

Explain why the refraction of the sound wave is different from the refraction of the light wave in Figure 10.

(3)

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(Total for question = 3 marks)

Q2.

(i) Figure 2 shows a student sitting on the shore of a lake watching ripples on the surface of the water moving past a toy boat.

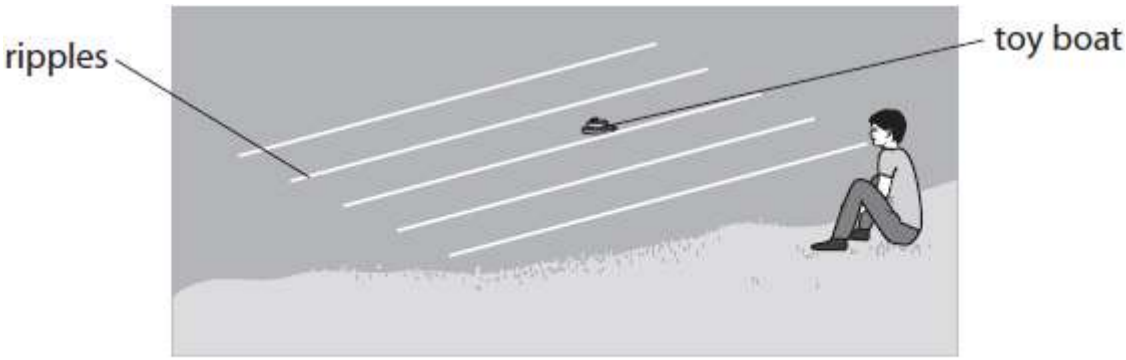

Figure 2

The student has a stopwatch.
Describe how the student could determine the frequency of the ripples on the lake.

(3)

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(ii) The speed of a water wave is 1.5 m/s.

The frequency of the wave is 0.70 Hz.
Calculate the wavelength of this wave.
Use the equation

$$v = f \times \lambda$$

(2)

wavelength = ... m

(iii) Water waves are transverse waves.

Describe the difference between transverse waves and longitudinal waves.

(2)

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(Total for question = 7 marks)

Q3.

The speed of light is 3.0×10^8 m/s.

The wavelength of yellow light is 5.8×10^{-7} m.

Calculate the frequency of yellow light.

State the unit.

Use the equation

$$\text{frequency} = \frac{\text{speed}}{\text{wavelength}}$$

(3)

frequency = ... unit .....

(Total for question = 3 marks)

Q4.

This picture shows Galileo holding a telescope.

(a) Explain the purpose of the eyepiece in a telescope.

(2)

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(b) Galileo drew pictures of his observations of Jupiter.
Nowadays we can take photographs.
Suggest how photographs would have helped Galileo.

(1)

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(c) The telescope collects light reflected from Jupiter.
The light has a frequency of 4.30×10^{14} Hz and a speed of 3.00×10^{8} m/s.
Calculate the wavelength of the light.

(3)

wavelength =..... m

*(d) Galileo's observations of the moons of Jupiter disproved the geocentric model.
However, these observations were not enough to prove the heliocentric model of the Solar System.
Explain why Galileo's observations disproved one model but were not enough to prove the other model.

(6)

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(Total for Question = 12 marks)

Q5.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

This question is about ultrasound.

Which of these is a frequency of ultrasound?

(1)

☐ **A** 2.3 Hz

☐ **B** 23 Hz

☐ **C** 2.3 kHz

☐ **D** 23 kHz

(Total for question = 1 mark)

Q6.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

This question is about both **infrasound** and **ultrasound**.

Which of these is a frequency of **infrasound**?

(1)

☐ **A** 2.2 Hz
☐ **B** 2.2 kHz
☐ **C** 22 Hz
☐ **D** 22 kHz

(Total for question = 1 mark)

Q7.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Figure 5 shows lightning from a thunderstorm over the sea.

(Source: ©National Geographic)

Figure 5

(i) Which row of the table is correct for the light and sound waves?

(1)

		Light waves	Sound waves
☐	A	longitudinal	longitudinal
☐	B	longitudinal	transverse
☐	C	transverse	longitudinal
☐	D	transverse	transverse

(ii) A person on a beach hears the sound of thunder 6.0 s after seeing the lightning.

The speed of the sound wave is 330 m/s and the speed of light in air is 3.0×10^8 m/s.
Estimate the distance between the lightning and the person.

(2)

distance = ... m

(Total for question = 3 marks)

Q8.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which row of the table shows two transverse waves?

(1)

☐ A	infrasound	infrared
☐ B	infrared	ultraviolet
☐ C	ultrasound	infrasound
☐ D	ultraviolet	ultrasound

(Total for question = 1 mark)

Q9.

Which row of the table shows two transverse waves?

(1)

☐ A	infrasound	infrared
☐ B	infrared	ultraviolet
☐ C	ultrasound	infrasound
☐ D	ultraviolet	ultrasound

(Total for question = 1 mark)

Q10.

Figure 11 shows a dolphin.

Figure 11

Dolphins emit sounds under water.
Scientists at four different research centres investigate the sounds made by dolphins.
Each centre chooses one of their dolphins and analyses the frequencies of the sounds the dolphin makes.

Figure 12 shows the range of frequencies from four different dolphins, W, X, Y and Z.

dolphins	lowest frequency (Hz)	highest frequency (kHz)
W	200	92
X	100	120
Y	85	105
Z	90	130

Figure 12

(i) Which of the following statements is a correct deduction from this information?

(1)

☐ **A** all four dolphins emit infrasound and ultrasound

☐ **B** all four dolphins emit ultrasound

☐ **C** only some of the four dolphins emit infrasound and ultrasound

☐ **D** only some of the four dolphins emit ultrasound

(ii) The frequency range of the sounds from dolphin W is 91.8 kHz.
Calculate the frequency range of the sounds from dolphin X.

(2)

range = ... kHz

(Total for question = 3 marks)

Q11.

Complete the sentence by putting a cross (☒) in the box next to your answer.

A bee's wings flap up and down with a frequency of 230 Hz.

This is a frequency of

(1)

☐ **A** infrasound

☐ **B** sonar

☐ **C** sound

☐ **D** ultrasound

Mark Scheme

Q1.

	Answer	Additional guidance	Mark
	an explanation linking: (refraction caused by) change in speed (1) sound speeds up (when entering water) (1) light slows down (when entering water) (1)	allow 3 marks for sound speeds up and light slows down (when entering water) allow 2 marks for sound speeds up (when entering water) OR light slows down (when entering water) if no other marks scored allow 1 mark for description of any speed change (when entering water) if no other marks scored allow 1 mark for wavelength changes ignore references to transverse, longitudinal.	**(3)** **AO1**

Q2.

Question number	Answer	Additional guidance	Mark
(i) CS1	a description to include count the number of waves(1) (arriving/passing a point) in a specific time(1) use frequency = $\frac{\text{number of waves}}{\text{time}}$ (1)	 ignore in one second count the number of waves in one second scores 2 marks (MP1 and MP3) find the time between one wave and the next scores 2 marks (MP1 and MP2)	(3) AO1

Question number	Answer	Additional guidance	Mark
(ii) CS1	substitution (1) $1.5 = 0.7 \times \lambda$ rearrangement and evaluation 2.1(4) m	 $\frac{1.5}{0.7}$ allow $\frac{0.7}{1.5}$ for 1 mark award full marks for correct answer without working. $\lambda = v/f$ scores 1 mark	(2) AO2

Question number	Answer	Additional guidance	Mark
(iii) CS1	A description to include: mention of oscillations/vibrations (1) EITHER transverse – (oscillations) perpendicular to direction of wave (travel) (1) OR longitudinal - (oscillations) in same direction as wave (travel) (1)	up and down OR side to side (movements) OR back and forth transverse movement up and down but longitudinal is side to side (1 mark only)	(2) AO1

Q3.

Question Number	Answer	Additional guidance	Mark
	substitution (1) $\frac{3.0 (\times 10^{8})}{5.8 (\times 10^{-7})}$		(3) AO 2 1
	evaluation (1) 5.2×10^{14}	answers that round to 5.2×10^{14} award 2 marks for a correct answer without working allow 1 mark for answers that round to 5.2 to any power of ten	
	unit (1) Hz	independent mark accept hz or s^{-1} or per sec(ond) or hertz accept kHz, MHz etc with correct power (10^{11} kHz, 10^{8} MHz)	

Q4.

Question Number	Answer	Acceptable answers	Mark
(a)	An explanation linking any two of: MP1 magnify (1) with one of: MP2 the (real) image from objective (lens) (1) MP3 to provide greater detail (1)	 enlarges / bigger ignore zooming the real image (in the telescope) / image at focal point ignore make it clearer inversion of image focuses image	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b)	One sensible suggestion, such as • made recording results {easier /quicker} (1) • results more convincing to other people (1) • photograph is to scale (1)	For example, don't have to keep looking through telescope would be (better) proof / evidence (i.e. between geo- and helio-centric models) (eg multiple photographs would prove movement/orbit of moons) (photograph) is more {accurate / precise / reliable}/ can measure (relative) separations of moons (from planet) better ignore more detail/clearer/zooming	**(1)**

Question Number	Answer	Acceptable answers	Mark
(c)	Substitution (1) $3.0 \times 10^8 = 4.3 \times 10^{14} \times \lambda$ Transposition (1) $(\lambda=)\ \frac{3.0 \times 10^8}{4.3 \times 10^{14}}$ Evaluation (1) 6.98×10^{-7} (m)	 Substitution and transposition in either order Ignore triangle correct answer no working = 3 power of ten error = 2 to at least 2sf (eg 7.0/6.97.....)x10^{-7} Ignore powers of 10 until evaluation	**(3)**

Question Number		Indicative content	Mark
QWC	*** (d)**	A description to include some of the following points description of models • geocentric • heliocentric description of one set of observations of Jupiter's moons explanation of how observation contradicts geocentric but does not prove the other NB beware that you do not reward repetitions of the question stem	**(6)**
Level	**0**	no rewardable material	
1	**1-2**	• a limited explanation of the geocentric **AND** the heliocentric model such as *geocentric model said everything orbited the Earth while the other was for everything going round the Sun.* **OR** clearly explains the one model and describes Galileo's observations *eg geocentric model said everything orbited the Earth but Galileo observed that Jupiter had moons going around it* • the answer communicates ideas using simple language and uses limited scientific terminology e.g. allow confusion between geocentric and heliocentric • spelling, punctuation and grammar are used with limited accuracy	
2	**3-4**	• a simple explanation of geocentric **AND** heliocentric models **AND** Galileo's observations of Jupiter's moons/explains heliocentric not proved e.g. *The geocentric model said everything orbited the Earth while the heliocentric was for everything orbiting the Sun. Galileo observed that Jupiter had moons orbiting around it.* • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately e.g. correct use geocentric and heliocentric • spelling, punctuation and grammar are used with some accuracy	
3	**5-6**	• a detailed explanation of geocentric **AND** heliocentric models **AND** the role of Galileo in providing evidence against the geocentric **BUT** not enough for the heliocentric such as *The geocentric model said everything orbited the Earth while the heliocentric was for everything orbiting the Sun. Galileo's observations that Jupiter had moons orbiting arou(nd it showed that the geocentric model was wrong* ***but not*** *that Jupiter or anything else went around the Sun.* • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately e.g. distinguishes between the necessary and insufficient conditions • spelling, punctuation and grammar are used with few errors	

Q5.

Question number	Answer	Mark
	D 23 kHz A,B and C are not correct because they are all below 20 kHz	**(1)** **AO2**

Q6.

Question Number	Answer	Mark
	The only correct answer is A 2.2 Hz ***B,C*** *and* ***D*** *are not correct because they are too high.*	**(1)** **AO1.1**

Q7.

Question number	Answer	Additional guidance	Marks
(i)	C transverse longitudinal	Multiple choice	(1)

Question number	Answer	Additional guidance	Marks
(ii)	Substitution (1) (Distance =) 330 x 6.0 Evaluation (1) 2000 (m)	award full marks for the correct answer without working 1980(m) 1.98 km 2.0 km	(2)

Q8.

Question	Answer	Mark
	B infrared ultraviolet A incorrect as infrasound is a longitudinal wave C ultrasound and infrasound are both longitudinal waves D ultrasound is a longitudinal wave	**1** **AO1.1**

Q9.

Question	Answer	Mark
	B infrared ultraviolet A incorrect as infrasound is a longitudinal wave C ultrasound and infrasound are both longitudinal waves D ultrasound is a longitudinal wave	**1** **AO1.1**

Q10.

Question Number	Answer	Mark
(i)	B all four dolphins emit ultrasound	**(1)**

Question Number	Answer	Additional guidance	Mark
(ii)	calculation including conversion of lower frequency into kHz 120 – 0.1 (1) evaluation of range 119.9 (kHz) (1)	allow 1 mark for evaluation arising from failure to correctly convert Hz into kHz (e.g. 119 kHz) full marks will be awarded for correct numerical answer without working	**(2)**

Q11.

Question Number	Answer	Acceptable answers	Mark
	C sound		**(1)**